K-FIELD LITERATURE REVIEW

Drop-Tower Tests of Rotating Gyroscopes, Flywheels, and Oscillating Test Masses

Source-verified experimental parameters, K-Field maximum-envelope calculations, observer-indexed temporal analysis, and a complete Cell3 geometric reference.

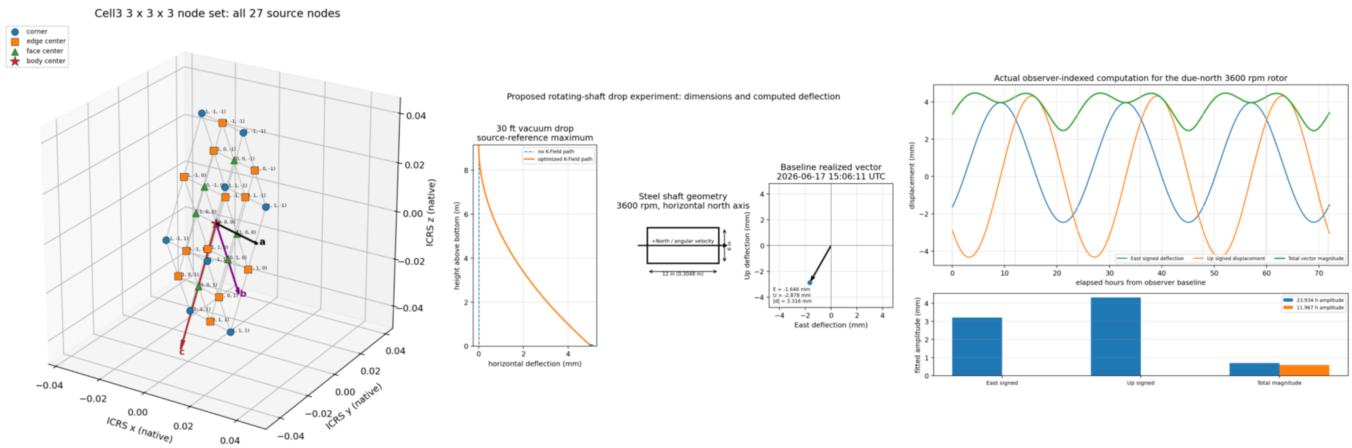

Figure 1. Composite overview generated from the source Cell3 geometry, the defined 30 ft rotating-shaft drop configuration, and the observer-indexed 72-hour computation.

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Abstract

This reference examines publicly available drop-tower and precision-measurement research involving rotating gyroscopes, flywheels, and related active test masses. The analysis applies a strict source-completeness rule: a K-Field acceleration or displacement is calculated only when the public source supplies the required rotation rate, active rotating mass and geometry, total accelerated mass when the rotor is internal to a larger falling assembly, and a fall duration or height. When exact epoch and laboratory orientation are absent, the calculation is explicitly limited to a maximum vector-alignment envelope. Papers lacking the required data are documented but not assigned estimated K-Field values.

Three source-complete configurations support direct calculations: the Luo free-fall gyroscope experiment, the improved Zhou experiment, and the Han concentric rotating-mass drop-tower design. In all three, the rotor spin axis and the precision measurement axis were vertical. Under the present K-Field cross-product law, the force is perpendicular to the spin axis; therefore, the ideal response in each published vertical measurement channel is exactly zero even though a potentially larger transverse envelope can be calculated. The reviewed literature does not contain a confirmed, time-tagged approximately 12-hour lateral-velocity or lateral-deflection residual. One horizontal-axis report explicitly states that daily dependence was not observed, but it does not publish the inputs or time series needed for a K-Field reconstruction. No completed mechanical oscillating-test-mass drop experiment was located that publishes oscillator mass, amplitude, frequency, phase, drop interval, and two-axis trajectory data; therefore no oscillating-mass K-Field value is assigned.

For the defined 6 inch diameter by 12 inch steel shaft rotating at 3,600 rpm in a 30 ft vacuum drop, the source-reference maximum lateral deflection is 5.029 mm. After applying the controlling observer state, location, epoch, and fixed due-north shaft orientation, the realized baseline vector is -1.646 mm east and -2.878 mm up, with total magnitude 3.316 mm. A 72-hour source-model computation shows that signed components are dominated by the 23.934-hour sidereal term, while the unsigned vector magnitude contains a material 11.967-hour semisidereal component. This distinction defines the correct search target for future drop data.

1. Scope, Evidence Rule, and Computational Boundary

The objective is to determine whether earlier rotating-body drop experiments resemble the proposed K-Field drop-tower test and whether their published residuals contain an unresolved lateral-velocity or trajectory-deflection component. The review separates four categories: source-reported apparatus data, source-reported measurements, K-Field calculations made from complete public inputs, and non-computable cases where an essential input is absent.

Mandatory input rule. No K-Field acceleration or displacement is assigned unless the source provides: (1) rotor RPM or angular rate; (2) active rotating mass and rotor geometry; (3) the total accelerated mass when the rotor is only part of a falling container; (4) fall time or fall height; and (5) enough axis information to determine whether the published measurement channel can see the cross-product response. Exact epoch and site orientation are additionally required for a realized vector; otherwise only the maximum orientation envelope is permitted.

```
F_K = (kappa_F * m_active * omega / 2) * (n_hat x g_vector)
```

```
a_assembly = F_K / m_total = q_g * omega * (m_active / m_total) * sin(theta)
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```
d_K = 0.5 * a_K * T^2 or, for an ideal drop from rest, d_K = a_K * h / g0
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For a freely falling object whose entire mass is the rotating active mass, m_active/m_total equals one and mass cancels from acceleration. For a rotor enclosed in a nonrotating frame, the active-mass fraction must be carried explicitly. This distinction is essential for the Luo and Zhou experiments.

2. Observer-State Baseline and Directional Gravity Handling

Input	Controlling value	Use
Reference frame	SGA-2020 / ICRS / J2000	All inertial vectors and Cell3 geometry
Observer epoch	2026-06-17T15:06:11Z	Calibration time for observer velocity
Laboratory location	42.129 deg, -83.144 deg, 0 m	Local ENU-to-ICRS rotation and surface velocity
Observer speed	341.071130 km/s	Magnitude of baseline observer-state velocity
Observer RA, Dec	167.143423 deg, -7.655565 deg	Direction of baseline observer-state velocity
CMB closure unit vector	(-0.97077086, 0.20734801, -0.12087493)	Fixed celestial reference direction
f_obs	1.447739	Observer-indexed model factor
beta_obs^2	1.555741579e-06	Observer-state scalar
zeta_R	(0.320892, 0.374911, 0.838466)	Observer orientation state

The observer velocity is reconstructed from the stated speed, right ascension, and declination. The annual orbital and site-rotation terms are calibrated so that their sum exactly reproduces the supplied observer vector at the controlling epoch. The phase-response vector is then g_vector(t) = G_phase V_lab(t). The local firing or rotor axis is transformed from east-north-up coordinates into ICRS/J2000 before the cross product is evaluated. Scalar G-derived rows are not pooled with non-directional residual families.

K-Field quantity	Value
kappa_F recommended	3.074050899702e-11
kappa_F interval	[3.042739944755e-11, 3.105361854649e-11]
Reference |g_vector|	930825.217270 m/s
q_g = kappa_F |g| / 2	1.430702048308e-05 N/(kg rad/s)
G_phase row 1	[2.9764106378, 1.2431507795, -2.0803600354]
G_phase row 2	[1.2431507795, 2.5922286635, -2.3178839937]
G_phase row 3	[-2.0803600354, -2.3178839937, 6.7735709133]

3. Proposed 30 ft Horizontal-Axis Rotor Experiment

Proposed rotating-shaft drop experiment: dimensions and computed deflection

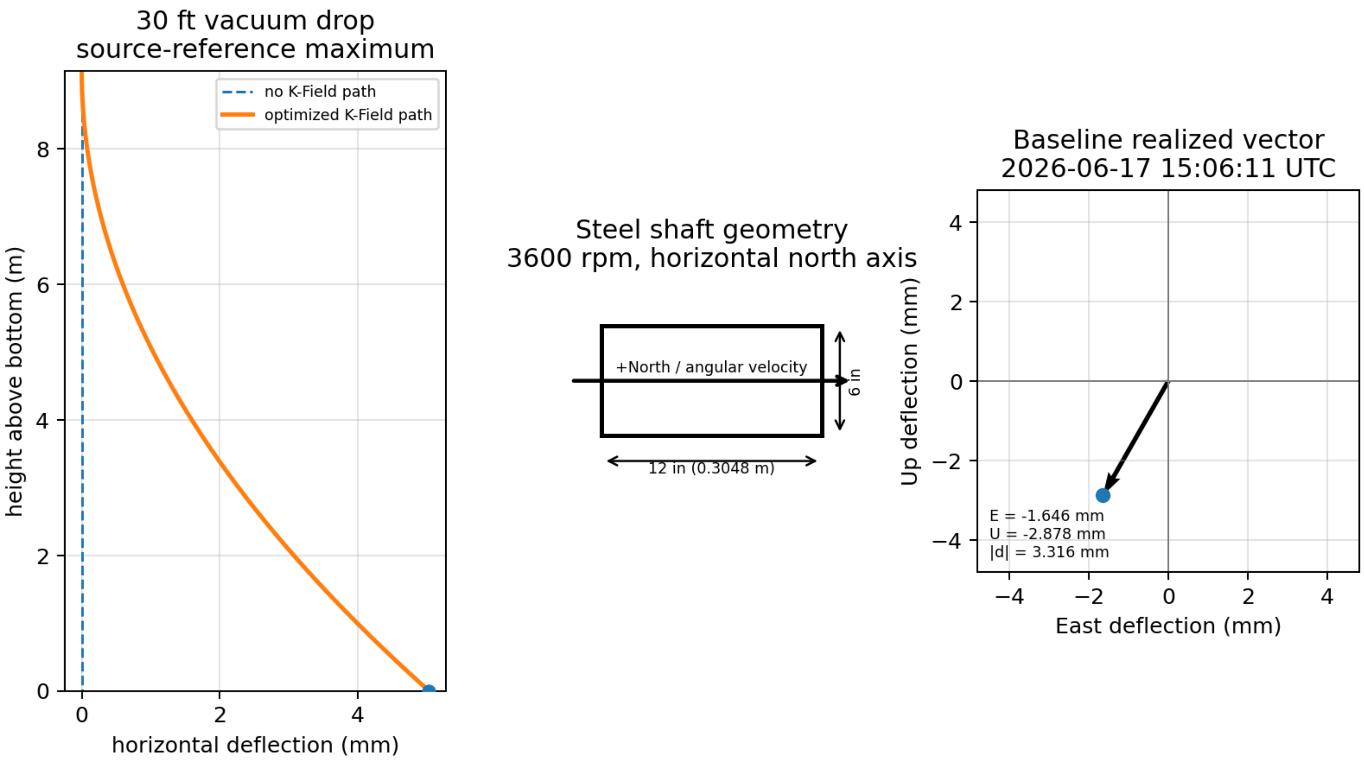

Figure 2. Dimensioned proposed apparatus and computed deflections. The tower path uses the exact 30 ft fall. The rotor inset uses the exact 6 inch diameter and 12 inch length. The baseline vector uses the controlling observer state and a due-north angular-velocity axis.

Quantity	Value	Derivation or condition
Diameter	6.000 in = 0.152400 m	User-defined
Length	12.000 in = 0.304800 m	User-defined
Working steel density	7850.0 kg/m^3	Used only to compute force and mass
Mass at working density	43.645999 kg	rho*pi*(D/2)^2*L
Spin rate	3600.0 rpm	User-defined
Angular rate	376.991118 rad/s	2*pi*RPM/60
Drop height	30.0 ft = 9.144000 m	User-defined
Ideal fall time	1.365598 s	sqrt(2h/g0)
Source-reference maximum acceleration	0.005393620 m/s^2	q_g*omega
Source-reference maximum force	0.235410 N	m*a; density-dependent
Source-reference maximum deflection	5.029165 mm = 0.197999 in	a*h/g0
Final transverse velocity at maximum	0.007365514 m/s	a*T

The 5.029 mm value is the source-reference maximum using the fixed q_g normalization. It is independent of rotor mass because the whole freely falling shaft is the active rotating mass. The mass and assumed steel density affect the force but not the ideal center-of-mass acceleration. The maximum is realized only when the transformed field vector is perpendicular to the shaft axis and the cross product lies horizontally.

Observer-indexed result	Value
Baseline g-vector magnitude	825840.226823 m/s
Epoch-specific optimum deflection	4.461940 mm
Fixed due-north alignment sin(theta)	0.743106914
East deflection	-1.645717 mm
North deflection	-3.078798e-16 mm
Up displacement	-2.878450 mm
Total K displacement magnitude	3.315699 mm
Final East velocity	-2.410251 mm/s
Final Up velocity	-4.215663 mm/s

Clockwise rotation viewed from south toward north corresponds to an angular-velocity vector directed north. For a due-north axis, the cross product contains east and vertical components but no north component. At the controlling epoch the horizontal trajectory deflection is therefore 1.646 mm west; the additional downward component changes the vertical trajectory by 2.878 mm. The total K displacement vector is 3.316 mm.

4. Literature Survey Matrix

Study	Rotor data	Drop data	Precision channel	K computation	12-hour evidence
Hayasaka et al., Speculations in Science and Technology 20, 173 (1997), as summarized by Luo et al.	partial	none	fall acceleration	not computed	not analyzed
J. Luo et al., Phys. Rev. D 65, 042005 (2002).	complete	complete	vertical	maximum envelope only	not analyzed
Z. B. Zhou et al., Phys. Rev. D 66, 022002 (2002).	complete	complete	vertical	maximum envelope only	not analyzed
C. G. Shao et al., Commun. Theor. Phys. 39, 297-300 (2003).	partial	partial	trajectory theory	not computed	not analyzed
A. L. Dmitriev, E. M. Nikushchenko, and S. A. Bulgakova, arXiv:0907.2790 (2009).	partial	partial	fall acceleration	not computed	explicit no daily dependence
A. L. Dmitriev, arXiv:1101.4678 (2011).	partial	partial	fall acceleration	not computed	not analyzed
F. T. Han et al., Chinese Physics Letters 34, 100701 (2017).	complete	complete	vertical	computed geometry complete	not analyzed
T. Y. Liu et al., Scientific Reports 6, 31632 (2016).	none	none	weight/residual	not applicable no rotor	not analyzed
J. Min et al., npj Microgravity 7, 25 (2021).	none	none	weight/residual	not applicable no rotor	not analyzed
T. J. Quinn and A. Picard, Nature 343, 732-735 (1990).	available	not a drop	weight/residual	not computed static	not analyzed
I. L. Lorincz and M. Tajmar, arXiv:1506.02689 (2015).	available	not a drop	weight/residual	not computed static	not analyzed
J. M. Nitschke and P. A. Wilmarth, Phys. Rev. Lett. 64, 2115-2116 (1990).	available	not a drop	weight/residual	not computed static	not analyzed

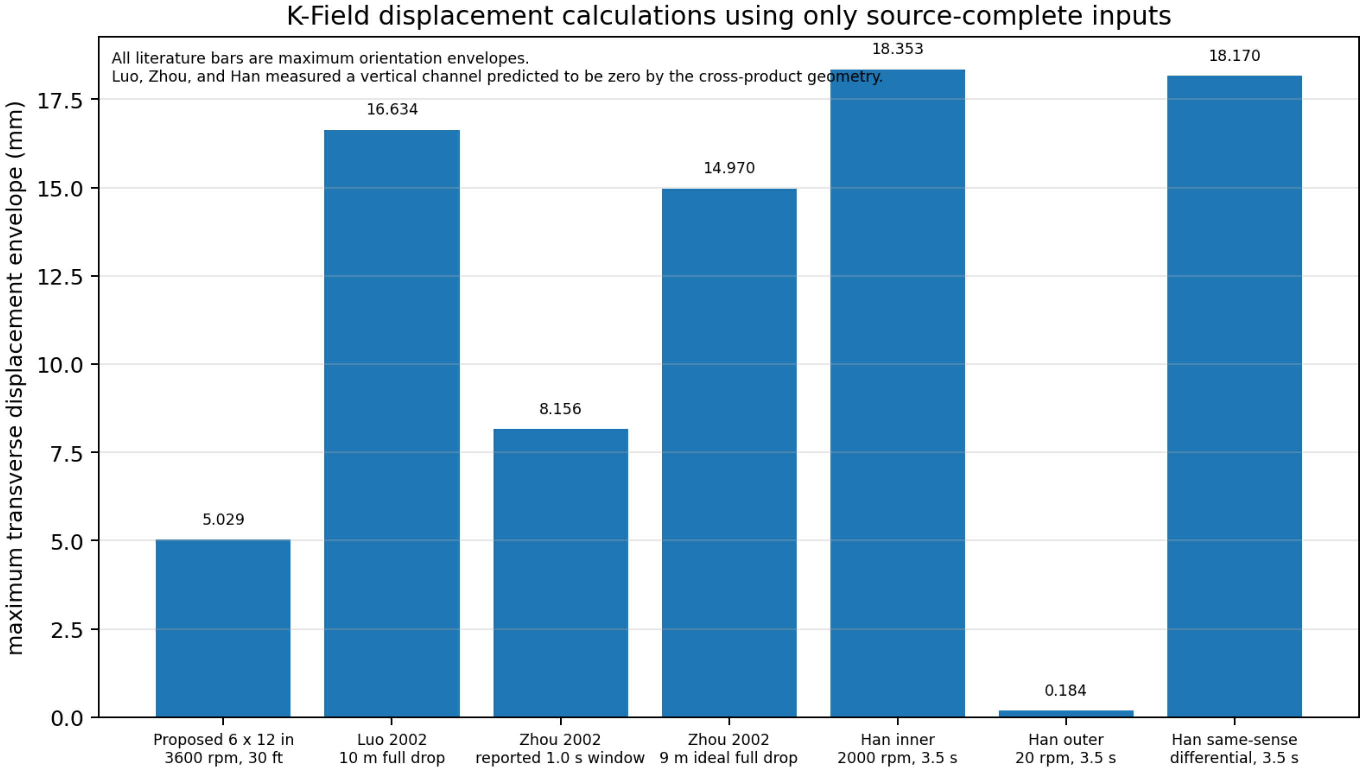

Figure 3. Maximum transverse displacement envelopes computed only for configurations with source-complete inputs. The reported precision channels for Luo, Zhou, and Han are vertical and therefore orthogonal to the calculated K-Field response.

5. Source-by-Source Review and K-Field Reconstruction

5.1 Hayasaka Free-Fall Gyroscope Report (1997)

Luo et al. summarize a single-free-fall experiment in which a gyroscope at 18,000 rpm was compared across rotation senses by laser timing. Their summary reports a relative fall-acceleration reduction of approximately 10^-4 for one rotation sense and no comparable change for the opposite sense [1]. Luo et al. also tabulate a 175 g rotor with a 5.8 cm diameter. The accessible reviewed source does not provide the complete falling assembly, a source-verified fall distance or duration, exact timestamps, or sufficient axis data to reconstruct the realized K-Field vector.

K-Field disposition: no force-to-assembly acceleration and no displacement are calculated. Importing an unverified drop height or assuming that the rotor mass equals the entire falling mass would violate the source-completeness rule.

5.2 Luo Double Free-Fall Interferometer (2002)

The Luo apparatus used two 12 m vacuum tubes. Each falling test mass contained a 420.0 +/- 2.5 g steel gyroscope, a 76.4 +/- 0.4 g corner-cube reflector, and a 159.4 +/- 0.9 g aluminum frame. The rotor was approximately 55 mm in diameter, 32 mm high, and operated at 17,000 +/- 200 rpm. The interferometer continuously measured differential vertical displacement. The paper reports no differential vertical acceleration at the relative 2 x 10^-6 level [1].

Computed quantity	Value
Active rotor mass	0.4200 kg
Total falling assembly mass	0.6558 kg
Active mass fraction	0.640439158
Angular rate	1780.235837 rad/s
Maximum force on rotor	0.010697346 N

Computed quantity	Value
Maximum transverse assembly acceleration	0.016311902 m/s^2
Height-derived time for 10 m	1.428087 s
Maximum transverse displacement over 10 m	16.633511 mm
Predicted ideal vertical-channel acceleration	0 m/s^2

The zero vertical-channel result follows directly from geometry. The rotor spin axis was vertical, so n_hat x g_vector is horizontal. The paper also states that the interferometer was nearly insensitive to horizontal motion and inferred a horizontal velocity bound from beam intensity rather than direct precision tracking. Exact timestamps are not public, so 16.634 mm is a maximum transverse envelope, not a reconstructed realized deflection.

5.3 Zhou Improved Double Free-Fall Experiment (2002)

The improved Zhou experiment retained the 420 g, 55 mm by 32 mm steel rotor and 17,000 rpm speed, improved the vacuum to 2 mPa, moved the pump, and extended the usable data from approximately 0.2 s to an effective 9 m and 1.0 s interval. The reported differential results were 0.48 microGal for nonrotating versus left rotation and -110 microGal for nonrotating versus right rotation, consistent with a total rotating-system uncertainty of 160 microGal. Rotor-frequency reflector vibration was approximately 0.1 micrometre, and slow frame motion associated with friction coupling occurred near 1.6 Hz [2].

Computed quantity	Value
Maximum force on active rotor	0.010697346 N
Maximum transverse assembly acceleration	0.016311902 m/s^2
Maximum transverse displacement over reported 1.0 s	8.155951 mm
Maximum transverse displacement for full ideal 9 m drop	14.970160 mm
Predicted ideal vertical-channel acceleration	0 m/s^2

The 1.0 s value is the source-controlled measurement-window displacement. The 9 m value is separately identified as the ideal full-height value obtained from h/g scaling. Neither is a prediction for the published vertical residual because that channel is orthogonal to the K-Field force under the stated axis geometry.

5.4 Shao Earth-Rotation Trajectory Analysis (2003)

Shao et al. analyzed long-distance free-fall trajectories for a rotating ball and a nonrotating shell while explicitly including Earth rotation. Their paper uses a 40 s fall interval and reports an orbit difference of order 10^-9 cm under the authors' spin-spin interaction model [15]. The paper is relevant because it treats trajectory deflection rather than only vertical acceleration.

K-Field disposition: no current K-Field acceleration, velocity, or displacement is computed. The public source does not state the rotating ball RPM, rotating mass, or rotor geometry. The paper's reported 10^-9 cm value belongs to its own spin-spin model and is not imported into the present force law.

5.5 Dmitriev Horizontal-Axis Free-Fall Container (2009)

This three-page report describes a closed falling container containing two coaxial horizontal-axis gyroscope rotors. Maximum speed was approximately 20,000 rpm, run time was 14-15 minutes, more than 200 photographs were processed, and the authors reported an average acceleration increase of 10 +/- 2 cm/s^2. The combined angular momentum at maximum same-direction rotation was reported as approximately 2.0 kg m^2/s. The paper states that N-S versus W-E orientation did not affect the result and that daily dependence was not observed [3].

K-Field disposition: no acceleration or displacement is computed. Rotor masses, rotor dimensions, complete container mass, and an unambiguous source-controlled fall interval are absent from this paper. The statement that daily dependence was not observed is retained as a source result, but the paper does not publish a time series or harmonic fit from which a 12-hour limit can be calculated.

5.6 Dmitriev Frequency-Dependence Study (2011)

The later study used a vacuum aviation rotor in an isolated container over 20-400 Hz, with a 30 mm fall path and approximately 40 ms readout. The container dimensions were 82 x 82 x 66 mm. The authors reported stochastic frequency dependence, narrow extrema most visible near 320 and 360 Hz, and a predominance of negative acceleration changes [4]. Measurements were spaced by 0.5-1 minute as the rotor ran down.

K-Field disposition: no acceleration or displacement is computed. Although RPM and fall path are known, the active rotor mass, rotor geometry, and total accelerated mass are not stated. The short spin-down sequence is not a 12-hour or sidereal experiment.

5.7 Han Concentric Rotating-Mass Drop-Tower Design (2017)

Han et al. provide the most complete published geometry for a purpose-built rotating-mass drop experiment. Two hollow aluminum cylinders are concentric in a differential electrostatic accelerometer. The inner mass is 27.1 g with radii 15.0 and 17.5 mm, length 39.92 mm, and nominal speed 2,000 rpm. The outer mass is 69.7 g with radii 25.0 and 27.5 mm, length 64.37 mm, and nominal speed 20 rpm. The Beijing drop time is 3.5 s. The sensitive x-axis and both spin axes are aligned with local vertical during the drop [5].

Quantity	Inner mass	Outer mass
Mass	27.100 g	69.700 g
Radii	15.0/17.5 mm	25.0/27.5 mm
Length	39.92 mm	64.37 mm
Speed	2000 rpm	20 rpm
Angular rate	209.439510 rad/s	2.094395 rad/s
Maximum transverse force	8.120394034e-05 N	2.088529388e-06 N
Maximum transverse acceleration	0.002996455 m/s^2	0.000029965 m/s^2
Maximum transverse displacement, 3.5 s	18.353289 mm	0.183533 mm
Axial moment of inertia	7.198437500e-06 kg m^2	4.813656250e-05 kg m^2
Angular momentum	1.507637224e-03 kg m^2/s	1.008169807e-04 kg m^2/s
Vertical sensitive-channel K acceleration	0	0

The same-sense speed-magnitude differential transverse acceleration is 0.002966491 m/s^2, corresponding to 18.169756 mm over 3.5 s. If the spin senses are opposite, the maximum separation envelope is 18.536822 mm. The article does not specify a completed drop result or the spin-sense pairing, so these are explicitly labeled geometric envelopes. The published precision channel remains zero under the current K-Field law because it is parallel to the spin axes.

5.8 Beijing Drop-Tower Residual Characterization (2016)

Liu et al. measured the facility itself rather than a rotating mass. The 83 m tower provided approximately 3.5 s of free fall. The double-capsule residual was better than 2 x 10^-4 g0. The paper identified air drag, capsule rotation, accelerometer recovery from 1 g saturation, and low-frequency tower-airflow structure as ordinary residual mechanisms [6]. No K-Field rotor calculation applies because no rotating active mass was used.

5.9 Taiji-1 Inertial-Sensor Substitute Drop Tests (2021)

Twenty three-axis tests produced measured residuals of approximately 58 micro-g along the fall direction and 13 micro-g horizontally during the most stable interval. An 87 micro-g acceleration transient and approximately 35 nm displacement vibration were associated with control-gain switching [7]. This is a modern three-axis residual benchmark, not a rotating-mass experiment. No K-Field rotor calculation applies.

5.10 Static Gyroscope-Weighing Studies

Quinn and Picard weighed a 330 g rotor while it spun down from 8,000 rpm. Apparent mass changes became insignificant after correction for friction torque and temperature [8]. Nitschke and Wilmarth reported no weight change from 0 to 22,000 rpm within -0.025 +/- 0.07 mg [10]. Lorincz and Tajmar tested vertical and horizontal orientations and showed that precession, vibration, and balance resonances can create apparent mass changes; their final relative bound was 2.6 x 10^-6 [9]. These experiments are directly relevant to systematic-error control but do not define a free-flight time, so no lateral drop displacement is computed.

5.11 Oscillating-Test-Mass Search Result

No completed public drop-tower experiment was located that combines a mechanically oscillating test mass with published oscillator mass, oscillation amplitude, frequency, phase, free-fall duration, and direct lateral position or velocity tracking. Dmitriev (2011) recommends ballistic gravimetry with rotating or oscillating test bodies, but the reported data are rotor spin-down data rather than a controlled oscillating-mass trajectory experiment [4].

K-Field disposition: no oscillating-mass acceleration or displacement is computed. The minimum source inputs required for such a calculation are absent from the located public record.

6. Residuals and Unidentified Components in the Public Record

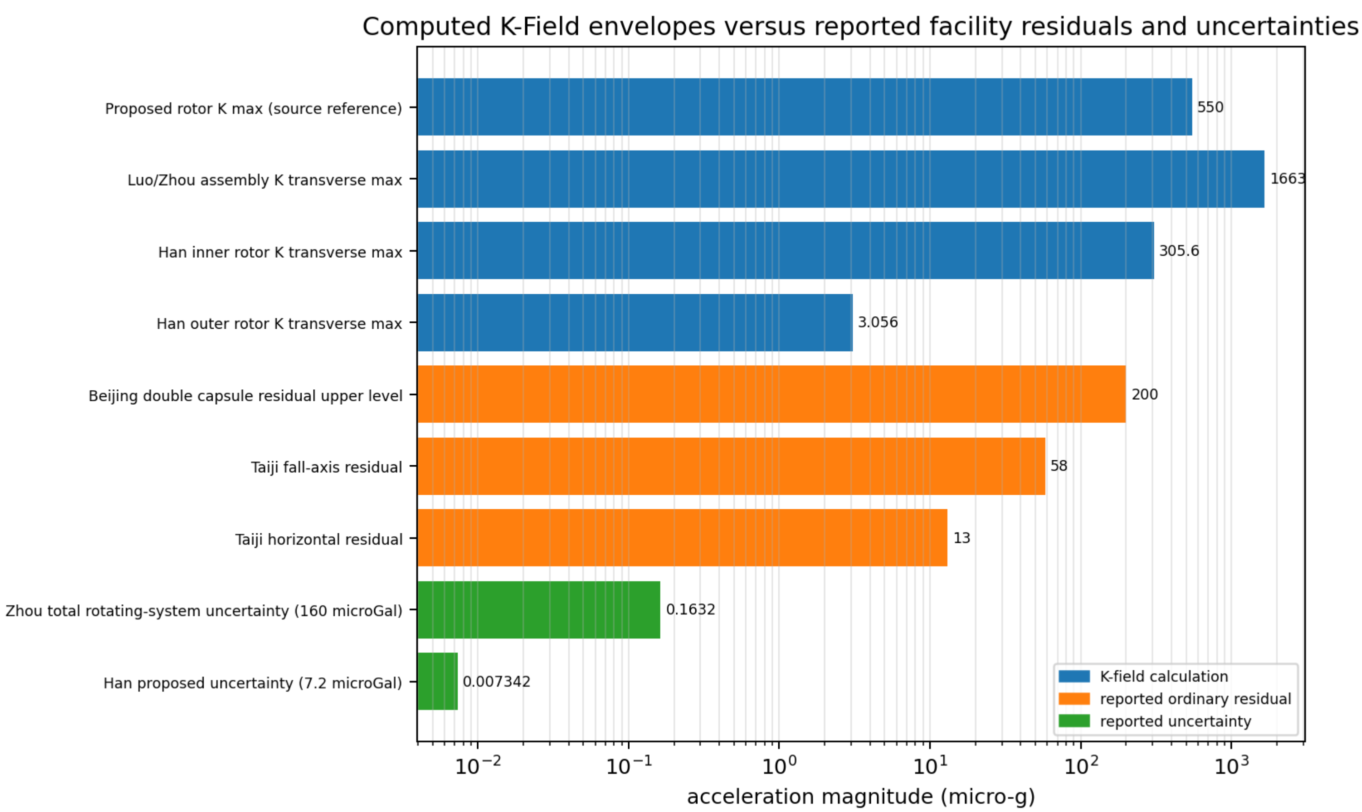

Figure 4. Acceleration-scale comparison. K-Field entries are computed maximum envelopes. Facility entries are source-reported ordinary residuals or uncertainties. The logarithmic axis prevents small precision terms from being hidden by larger drop-capsule residuals.

Residual mechanism	Source context	Observed or quantified behavior	Implication for lateral drop tests
Rotor-frequency vibration	Luo, Zhou, Lorincz-Tajmar	Reflector or balance motion at rotor frequency; Zhou reports order 0.1 micrometre modulation	Can move the optical reference without moving the center of mass
Slow frame rotation and friction coupling	Luo and Zhou	Approximately 1 Hz in Luo; approximately 1.6 Hz in Zhou; 130-150 microGal uncertainty in Zhou	Produces orientation change and apparent acceleration

Residual mechanism	Source context	Observed or quantified behavior	Implication for lateral drop tests
Horizontal release velocity / Coriolis	Luo and Zhou	Bounded indirectly; Zhou assigns <=54 microGal uncertainty	Must be measured as a vector rather than inferred from beam power
Outgassing and residual gas	Luo and Zhou	Asymmetric pump outgassing was a major limitation; pressure reduction improved results	Requires pump-off or symmetric pumping and direct pressure logging
Capsule translation and rotation	Liu and Han	Normal and tangential acceleration from off-axis sensor position	Measure six-degree-of-freedom capsule motion
Air drag and buoyancy	Liu and Dmitriev 2011	Low-frequency residual and height-dependent disturbances	Vacuum or double-capsule operation is required
Electrostatic switching	Taiji-1	87 micro-g transient and about 35 nm displacement vibration	Exclude switching windows or model the control impulse
Thermal and friction torque	Quinn-Picard; Lorincz-Tajmar	Apparent weight effects reduced after correction	Spin-reversal sequences must balance thermal history

The reviewed papers contain many residuals, but none provides a confirmed unexplained lateral velocity or displacement with a demonstrated approximately 12-hour recurrence. The strongest nonzero horizontal-axis papers do not publish enough apparatus data for a K-Field reconstruction. The strongest controlled vertical-axis free-fall papers measured a channel that the current K-Field geometry predicts to be zero.

7. Sidereal and Semisidereal Modulation

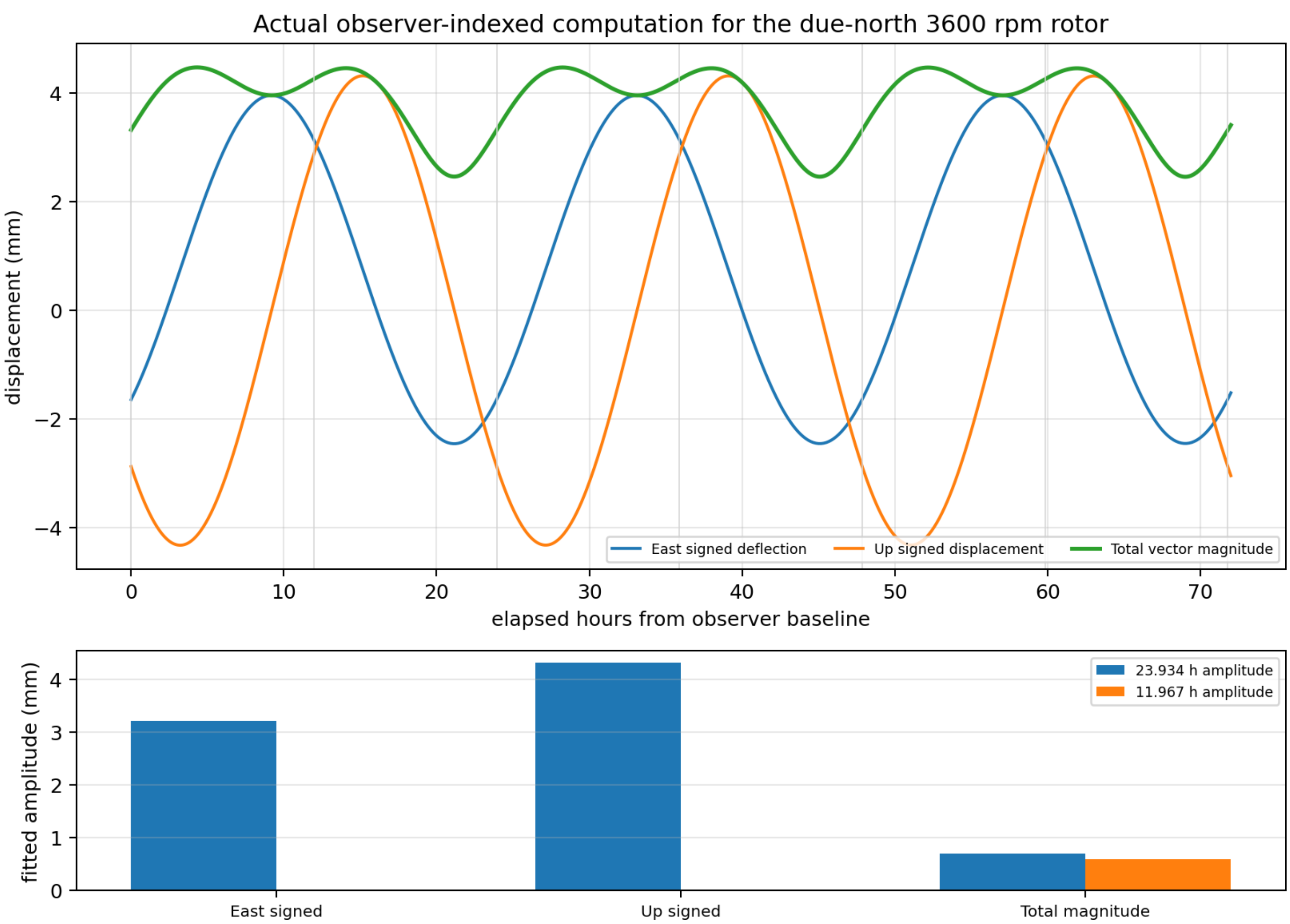

Figure 5. Seventy-two-hour computation using the controlling observer baseline, the due-north 3,600 rpm shaft axis, and the 30 ft drop. Vertical guide lines occur every half sidereal day. Signed components and total magnitude have different harmonic content.

Computed observable	Mean	23.934 h amplitude	11.967 h amplitude
East signed deflection (mm)	0.745868	3.20347	0.00241
Up signed displacement (mm)	-0.005797	4.316541	0.002638
Total displacement magnitude (mm)	3.82075	0.684369	0.585188
East final velocity (mm/s)	1.092368	4.691675	0.00353
Up final velocity (mm/s)	-0.008491	6.321835	0.003863
Total final-velocity magnitude (mm/s)	5.595719	1.002299	0.857043

The actual model computation does not support treating every component as a simple 12-hour sinusoid. Signed east and up components are overwhelmingly sidereal, with periods near 23 h 56 min. The total unsigned vector magnitude contains both sidereal and semisidereal content because a vector reversal after approximately half a sidereal day can preserve magnitude while changing sign. At this baseline, the fitted semisidereal amplitude of total deflection is 0.585 mm, while the signed east semisidereal amplitude is only 0.0024 mm.

Across the 72-hour computation, east deflection ranges from -2.457 to +3.952 mm, up displacement ranges from -4.325 to +4.312 mm, and total magnitude ranges from 2.453 to 4.467 mm. The epoch-specific optimum envelope ranges from 4.448 to 4.467 mm.

An observed approximately 12-hour laboratory cycle is not unique to K-Field geometry. NOAA identifies solar and lunar semidiurnal tidal constituents, and a complete experiment must record local gravity, tower tilt, temperature, pressure, and release conditions. The decisive discriminator is vector and spin parity: the K-Field term reverses with rotor spin, whereas most tidal, thermal, and structural terms do not.

```
d_spin_odd(t) = [d(+omega,t) - d(-omega,t)] / 2
```

8. Experimental Architecture for a Decisive Drop Test

Requirement	Implementation
Measured channels	Direct two-axis horizontal position plus vertical trajectory; derive final horizontal velocity from time-resolved data
Rotor states	Randomized +RPM, -RPM, and zero-spin drops with matched thermal and release histories
Axis states	Due north, due east, and selected optimized orientations; record axis attitude continuously
Time basis	UTC, local mean/apparent sidereal time, and observer-state epoch retained for every drop
Duration	At least 30 days for separation of solar and lunar semidiurnal backgrounds; longer duration improves annual/orbital separation
Release diagnostics	Initial horizontal velocity, release impulse, capsule angular velocity, and angular acceleration
Environmental diagnostics	Pressure, temperature, magnetic field, electrostatic charge, tower tilt, and local gravity
Primary statistic	Projection of the spin-odd displacement vector onto the epoch-specific predicted K-Field direction
Secondary statistic	Simultaneous fitted amplitudes at the sidereal and semisidereal frequencies

The proposed signal level is large relative to modern horizontal residual benchmarks when maximum alignment is approached. However, high signal size does not eliminate the need for vector readout. The prior literature demonstrates that rotor vibration and frame precession can move optical references, and that a vertical-only instrument can be blind to the current K-Field response. The new experiment should therefore be designed from the start as a three-dimensional trajectory measurement rather than a refined vertical gravimeter.

9. Findings

Finding	Result
Closest controlled free-fall precedents	Luo 2002 and Zhou 2002 provide complete rotor and assembly inputs, but measure a vertical channel predicted to be zero by the present K-Field cross product.
Closest horizontal-axis reports	Dmitriev 2009 and 2011 report nonzero acceleration structures but omit required rotor/assembly data; no K-Field acceleration or displacement is assigned.
Purpose-built rotating-mass drop design	Han 2017 provides complete mass, geometry, RPM, and duration. Its predicted transverse envelopes are calculable, while the published vertical sensitive channel remains zero.
Approximately 12-hour residual in literature	No confirmed public rotor-linked lateral velocity or deflection series was found. Dmitriev 2009 explicitly reports no daily dependence.
Oscillating test bodies	No source-complete completed drop-tower trajectory experiment was located; no oscillating-mass K-Field value is assigned.
Computed periodic structure for proposed experiment	Signed components are principally sidereal; total magnitude contains a material semisidereal component.
Proposed 3,600 rpm, 30 ft maximum	5.029 mm source-reference maximum; 4.462 mm epoch-specific optimum at the baseline; 3.316 mm total for the fixed due-north axis at the baseline epoch.

Central conclusion. The public record does not contain a clean prior test of the proposed observable: spin-odd, time-tagged, two-axis lateral deflection of a horizontal-axis rotor in free fall. The existing null experiments largely constrain vertical acceleration for vertical spin axes. The horizontal-axis nonzero reports are incomplete for quantitative K-Field reconstruction. The proposed 30 ft test therefore occupies a distinct and experimentally accessible measurement channel.

Appendix A. Cell3 Base Cell, Planes, and All Nodes

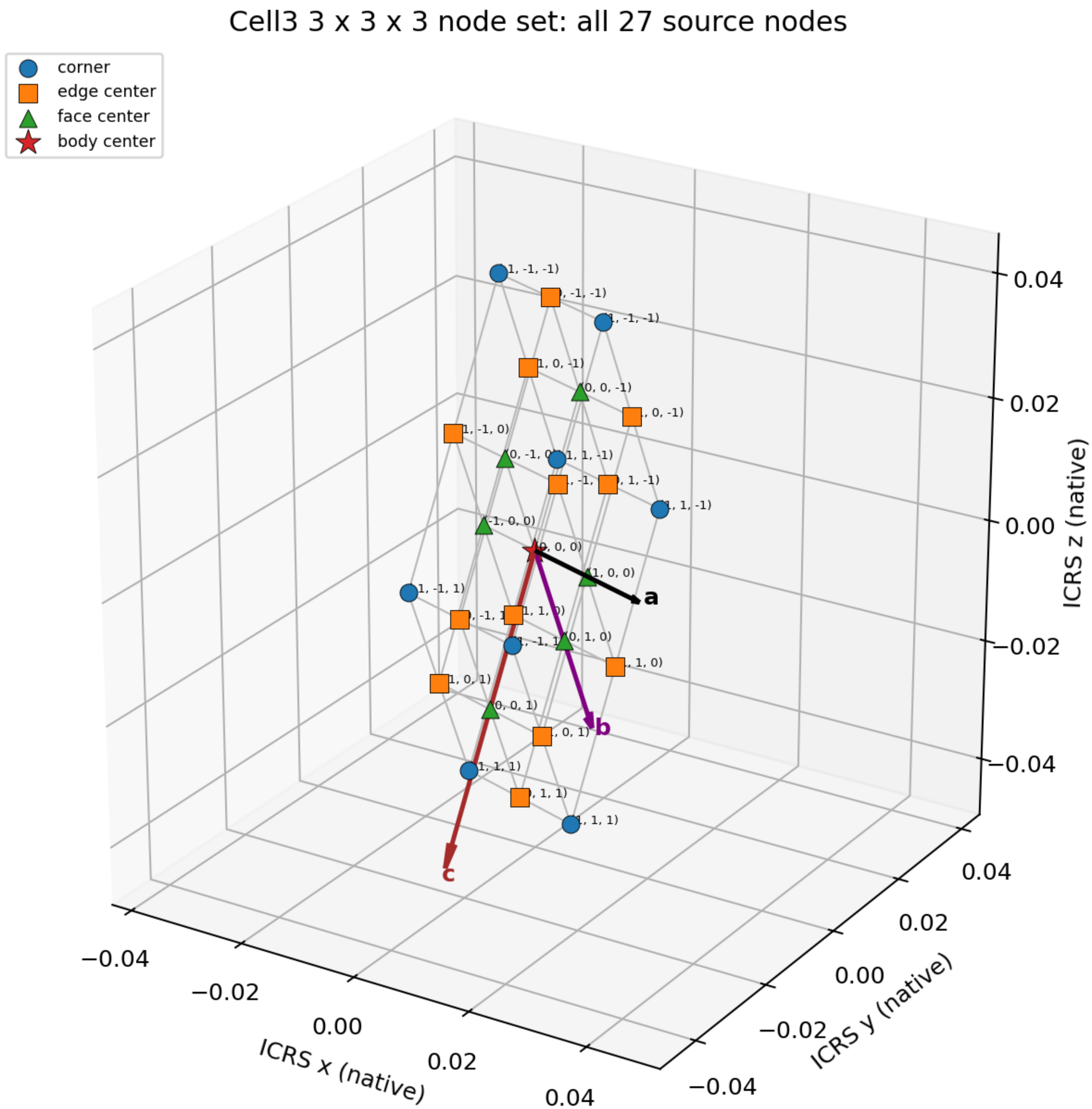

Figure A1. All 27 Cell3 nodes from the machine-readable source. Node coordinates are plotted in native ICRS units. Edges connect adjacent integer-index nodes, and the full basis vectors a, b, and c are shown.

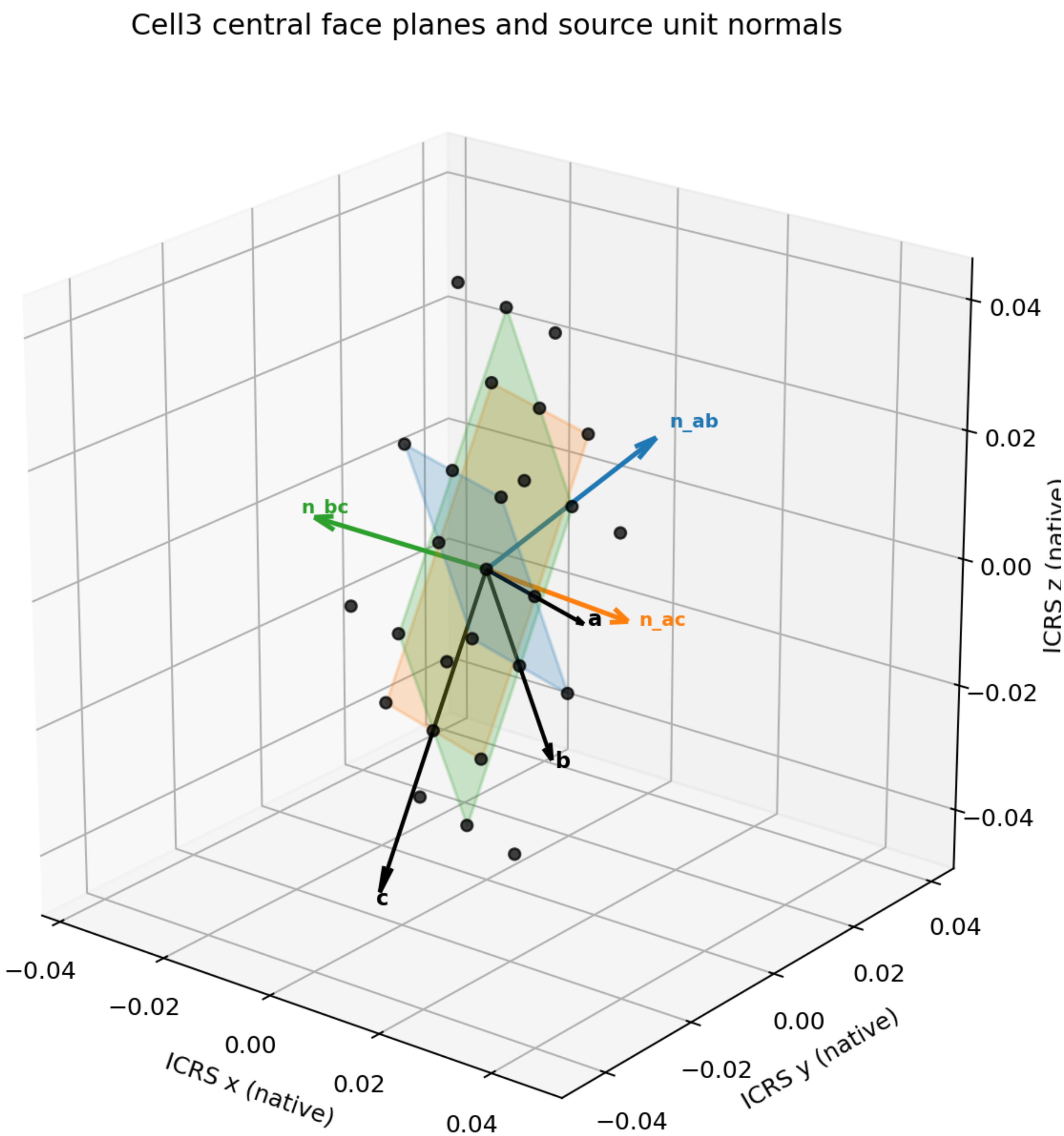

Figure A2. Central ab, ac, and bc face planes through the body center, using source basis vectors and source unit normals.

Vector	x native	y native	z native	Magnitude
a	0.020381438517	-0.003416414496	-0.002332965917	0.020797058782
b	0.002198745968	0.012902170259	-0.034066501536	0.036494205131
c	-0.000630379485	-0.024257722017	-0.042924354571	0.0493085659

Primitive volume: 2.842066937809649e-05 native^3.

Face	Area native^2	Included angle (deg)	Unit normal x	Unit normal y	Unit normal z
ab	0.000754721789673	83.933483012313	0.194092200808	0.913177168821	0.358379234791

Face	Area native^2	Included angle (deg)	Unit normal x	Unit normal y	Unit normal z
ac	0.001011255757688	80.448130018338	0.089052594102	0.866576769291	-0.491033948326
bc	0.001385784431733	50.363231813756	-0.995965190309	0.083602162507	-0.032619290564

Index (i,j,k)	x native	y native	z native	Node class	lambda sum
(-1, -1, -1)	-0.01097490249996	0.00738598312742	0.03966191101194	corner	-3
(-1, -1, 0)	-0.01129009224263	-0.00474287788114	0.01819973372639	edge center	-2
(-1, -1, 1)	-0.01160528198531	-0.0168717388897	-0.00326244355915	corner	-1
(-1, 0, -1)	-0.0098755295158	0.01383706825677	0.02262866024391	edge center	-2
(-1, 0, 0)	-0.01019071925847	0.00170820724821	0.00116648295836	face center	-1
(-1, 0, 1)	-0.01050590900115	-0.01042065376036	-0.02029569432718	edge center	0
(-1, 1, -1)	-0.00877615653164	0.02028815338612	0.00559540947588	corner	-1
(-1, 1, 0)	-0.00909134627431	0.00815929237755	-0.01586676780966	edge center	0
(-1, 1, 1)	-0.00940653601699	-0.00396956863101	-0.03732894509521	corner	1
(0, -1, -1)	-0.00078418324149	0.00567777587922	0.03849542805357	edge center	-2
(0, -1, 0)	-0.00109937298416	-0.00645108512935	0.01703325076803	face center	-1
(0, -1, 1)	-0.00141456272684	-0.01857994613791	-0.00442892651752	edge center	0
(0, 0, -1)	0.00031518974267	0.01212886100856	0.02146217728555	face center	-1
(0, 0, 0)	0	0	0	body center	0
(0, 0, 1)	-0.00031518974267	-0.01212886100856	-0.02146217728555	face center	1
(0, 1, -1)	0.00141456272684	0.01857994613791	0.00442892651752	edge center	0
(0, 1, 0)	0.00109937298416	0.00645108512935	-0.01703325076803	face center	1
(0, 1, 1)	0.00078418324149	-0.00567777587922	-0.03849542805357	edge center	2
(1, -1, -1)	0.00940653601699	0.00396956863101	0.03732894509521	corner	-1
(1, -1, 0)	0.00909134627431	-0.00815929237755	0.01586676780966	edge center	0
(1, -1, 1)	0.00877615653164	-0.02028815338612	-0.00559540947588	corner	1
(1, 0, -1)	0.01050590900115	0.01042065376036	0.02029569432718	edge center	0
(1, 0, 0)	0.01019071925847	-0.00170820724821	-0.00116648295836	face center	1
(1, 0, 1)	0.0098755295158	-0.01383706825677	-0.02262866024391	edge center	2
(1, 1, -1)	0.01160528198531	0.0168717388897	0.00326244355915	corner	1
(1, 1, 0)	0.01129009224263	0.00474287788114	-0.01819973372639	edge center	2
(1, 1, 1)	0.01097490249996	-0.00738598312742	-0.03966191101194	corner	3

Appendix B. Source-Completeness Ledger

Study	Known source inputs	Missing inputs	Calculation status
Hayasaka et al., Speculations in Science and Technology 20, 173 (1997), as summarized by Luo et al.	rpm, rotor_mass_g, rotor_diameter_cm	complete falling-assembly mass, source-verified fall height or duration, exact timestamp	not computed
J. Luo et al., Phys. Rev. D 65, 042005 (2002).	active_rotor_mass_kg, rotor_diameter_m, rotor_height_m, ccr_mass_kg, frame_mass_kg, total_falling_mass_kg, rpm, evaluated_fall_height_m	none	maximum envelope only
Z. B. Zhou et al., Phys. Rev. D 66, 022002 (2002).	active_rotor_mass_kg, rotor_diameter_m, rotor_height_m, total_falling_mass_kg, rpm, effective_height_m, analysis_interval_s	none	maximum envelope only
C. G. Shao et al., Commun. Theor. Phys. 39, 297-300 (2003).	fall_duration_s	rotor RPM or angular rate, rotor mass, rotor geometry, total falling mass, exact epoch, laboratory location and axis orientation	not computed
A. L. Dmitriev, E. M. Nikushchenko, and S. A. Bulgakova, arXiv:0907.2790 (2009).	rpm_max, combined_angular_momentum_kg_m2_s	rotor masses, rotor dimensions, complete container mass, unambiguous fall height or duration, exact timestamp	not computed
A. L. Dmitriev, arXiv:1101.4678 (2011).	frequency_range_hz, rpm_max, fall_path_m, readout_s, container_dimensions_mm	rotor mass, rotor dimensions, complete falling-container mass, exact timestamps	not computed
F. T. Han et al., Chinese Physics Letters 34, 100701 (2017).	drop_duration_s, inner, outer	none	computed geometry complete
T. Y. Liu et al., Scientific Reports 6, 31632 (2016).	free_fall_duration_s, double_capsule_residual_upper_g	none	not applicable no rotor
J. Min et al., npj Microgravity 7, 25 (2021).	free_fall_duration_s, horizontal_residual_micro_g, fall_axis_residual_micro_g, mode_switch_peak_micro_g, mode_switch_displacement_nm	none	not applicable no rotor
T. J. Quinn and A. Picard, Nature 343, 732-735 (1990).	rotor_mass_kg, rpm_max	drop interval (not a drop experiment)	not computed static
I. L. Lorincz and M. Tajmar, arXiv:1506.02689 (2015).	rotor_mass_kg, rpm_max_above	source-complete rotor geometry, drop interval (not a drop experiment)	not computed static
J. M. Nitschke and P. A. Wilmarth, Phys. Rev. Lett. 64, 2115-2116 (1990).	rpm_max	drop interval (not a drop experiment)	not computed static

Appendix C. Representative Observer-Indexed Samples

Elapsed h	Epoch UTC	East d (mm)	Up d (mm)	|d| (mm)	East v (mm/s)	Up v (mm/s)	Optimum d (mm)
0	2026-06-17T15:06:11Z	-1.64572	-2.87845	3.3157	-2.41025	-4.21566	4.46194
6	2026-06-17T21:06:11Z	2.89255	-3.21671	4.32597	4.23632	-4.71106	4.46635
12	2026-06-18T03:06:11Z	3.11467	2.90094	4.25636	4.56162	4.24859	4.45621
18	2026-06-18T09:06:11Z	-1.42205	3.17496	3.47888	-2.08267	4.64992	4.45115
24	2026-06-18T15:06:11Z	-1.60491	-2.9351	3.34523	-2.35049	-4.29862	4.46061
30	2026-06-18T21:06:11Z	2.93526	-3.16256	4.3148	4.29886	-4.63176	4.46489
36	2026-06-19T03:06:11Z	3.07509	2.95706	4.2662	4.50366	4.3308	4.45471
42	2026-06-19T09:06:11Z	-1.46259	3.12043	3.4462	-2.14206	4.57006	4.44986
48	2026-06-19T15:06:11Z	-1.5634	-2.99081	3.37479	-2.2897	-4.38022	4.45944
54	2026-06-19T21:06:11Z	2.97733	-3.10754	4.30364	4.36048	-4.55119	4.46359
60	2026-06-20T03:06:11Z	3.03495	3.01225	4.27604	4.44486	4.41162	4.45337
66	2026-06-20T09:06:11Z	-1.50238	3.06505	3.41346	-2.20033	4.48895	4.44874
72	2026-06-20T15:06:11Z	-1.52119	-3.04558	3.40435	-2.22788	-4.46044	4.45843

The companion JSON contains the complete 5-minute series for all 72 hours.

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